2. d) accepts OH^- from water ,releasing proton.

$$B(OH)_3 + H_2O \rightleftharpoons [B(OH)_4]^- + H^+$$

3. b) B_3H_6

nido borane : B_nH_{4+n}

aracno borane : B_nH_{6+n}

B_3H_6 is not a borane

4. a) Aluminium

5. c) four

There are two 3c – $2e^-$ bonds i.e., the bonding in the bridges account for 4 electrons.

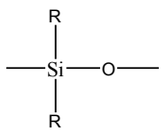
6. c) Lead

7. c) sp^2 hybridised

8. a) +4

Example : CH_{4+} in which the oxidation state of carbon is +4

9. d) $(SiO_4)^{4-}$

10. b) 

11. a) Me_3SiCl

12. d) dry ice

dry ice – solid CO_2 in which carbon is in sp hybridized state

13. a) Tetrahedral

14. d) Feldspar is a three dimensional silicate

15. a) A-b , B-1 , C-4 , D-3

16. d) Al,Cu,Mn,Mg

Al-95% , Cu-4% , Mn-0.5% , Mg-0.5%

17. a) Metal borides

18. a) Al < Ga < In < Tl

stability of +1 oxidation state increases down the group due to inert pair effect

UNIT-3

Choose the correct answer:

1. a) Nessler's reagent

2. d) ability to form $p\pi - p\pi$ bonds with itself

3. d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

4. b) P_4(white) and PH_3

5. a) H_3PO_3

6. a) H_3PO_3

7. b) 2

8. a) 6N

9. d) Both assertion and reason are wrong. The converse is true.

10. b) F_2

11. b) HF > HCl > HBr > HI

12. d) NeF_2

13. d) Kr

14. c) XeO_3

15. a) HI

16. d) $Cl_2 > Br_2 > F_2 > I_2$

17. d) $HClO < HClO_2 < HClO_3 < HClO_4$

18. c) $Cu(NO_3)_2$ and NO_2

Answers.indd 235 22-12-2022 19:40:01